

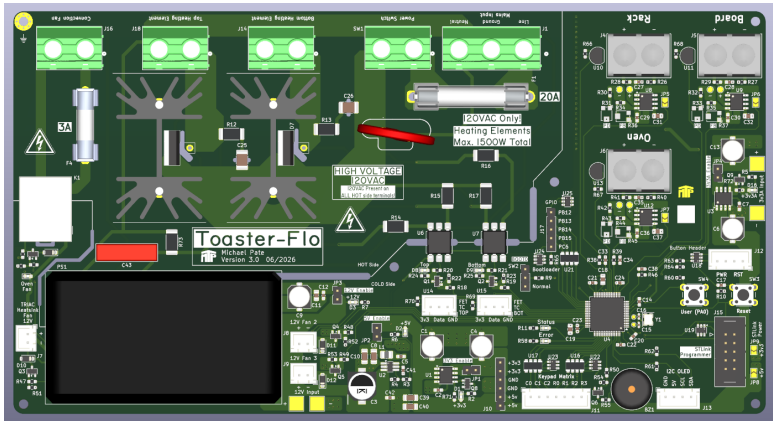
TF Main Board

Variant:

2026-07-29
Rev 2.0

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TOP VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

NOTES

Add a comment here

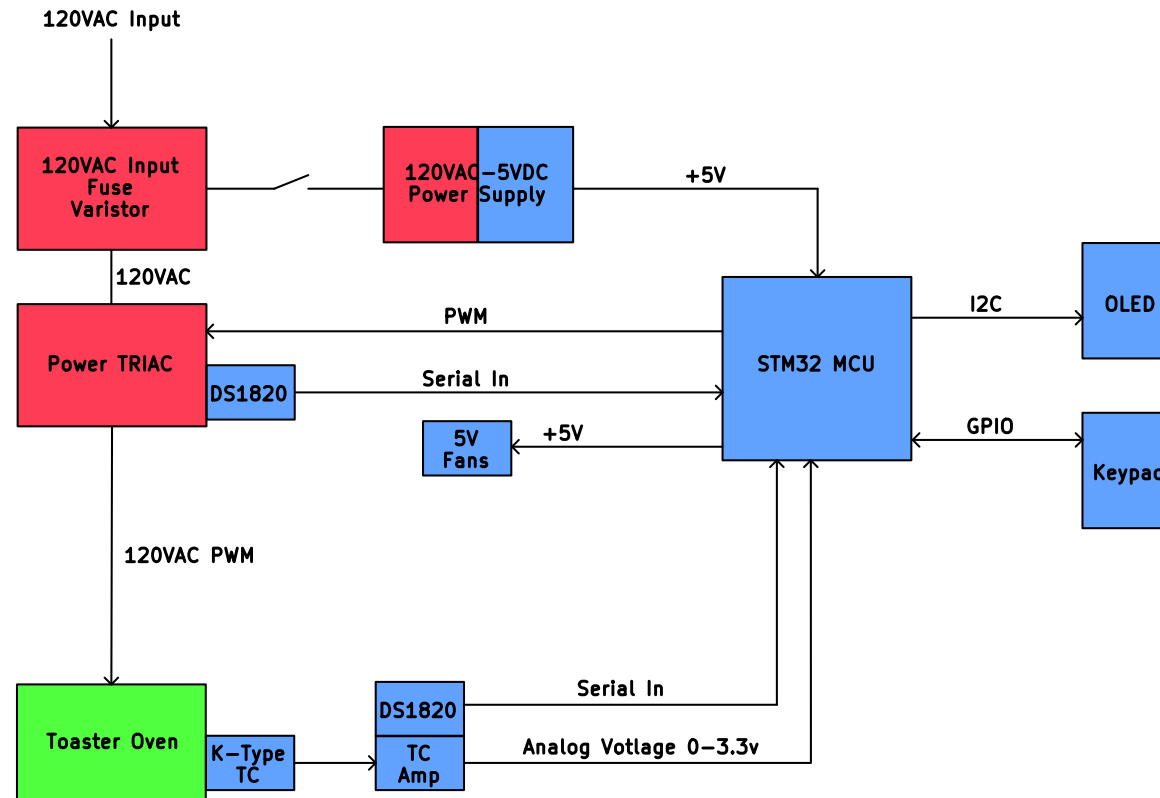
Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Release Date

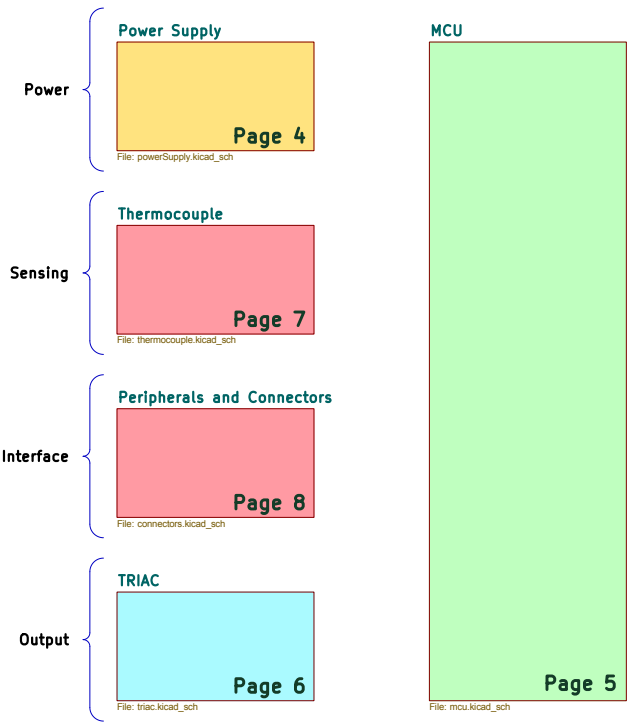
Comments:	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Cover Page	File Name: mainBoard.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /	Reviewer: Michael Pate		Size: A3	Sheet: 1 of 9

[2] Block Diagram



	Comments:		Company: MTP Engineering LLC		Variant:	
			Board Name: TF Main Board		Project Name: ToasterFlo	
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
	Sheet Path: /Block Diagram/			Reviewer: Michael Pate	Size: A3	Sheet: 2 of 9

[3] Project Architecture



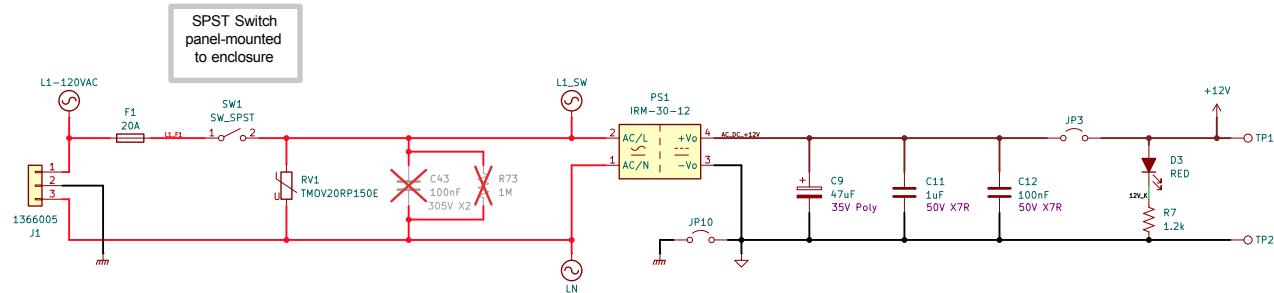
	Comments:		Company: MTP Engineering LLC		Variant:	
			Board Name: TF Main Board		Project Name: ToasterFlo	
	Sheet Title: Project Architecture		File Name: Project Architecture.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
	Sheet Path: /Project Architecture/			Reviewer: Michael Pate	Size: A3	Sheet: 3 of 9

[4] Power Supply

HIGH VOLTAGE:
Ensure appropriate isolation between AC and DC circuits!

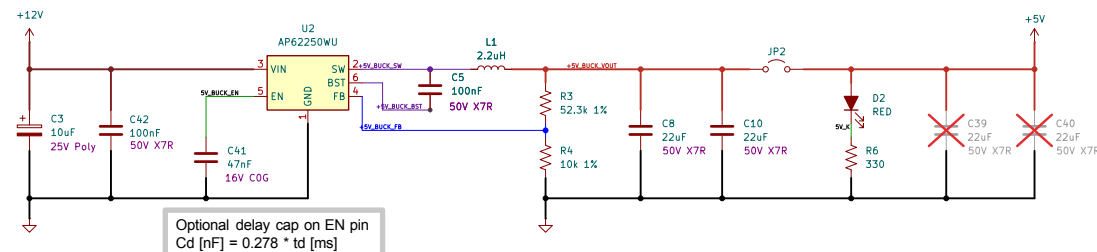
Air isolation gap underneath PS1

120VAC Only!
PCB clearances not designed for >120VAC



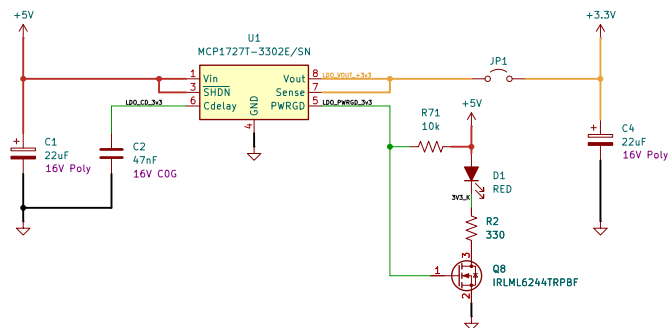
120VAC – 12VDC Supply

High Frequency! 1.3MHz
Care should be taken to route switching node for buck converter

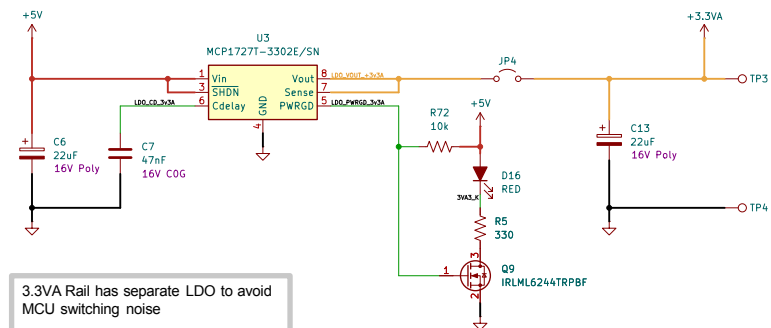


Optional delay cap on EN pin
 $C_d [nF] = 0.278 * t_d [ms]$

5VDC Buck Converter



3.3VDC LDO Digital



3.3VDC Rail has separate LDO to avoid MCU switching noise

3.3VDC LDO Analog

Use M2.5 mounting screws to secure PCB to housing

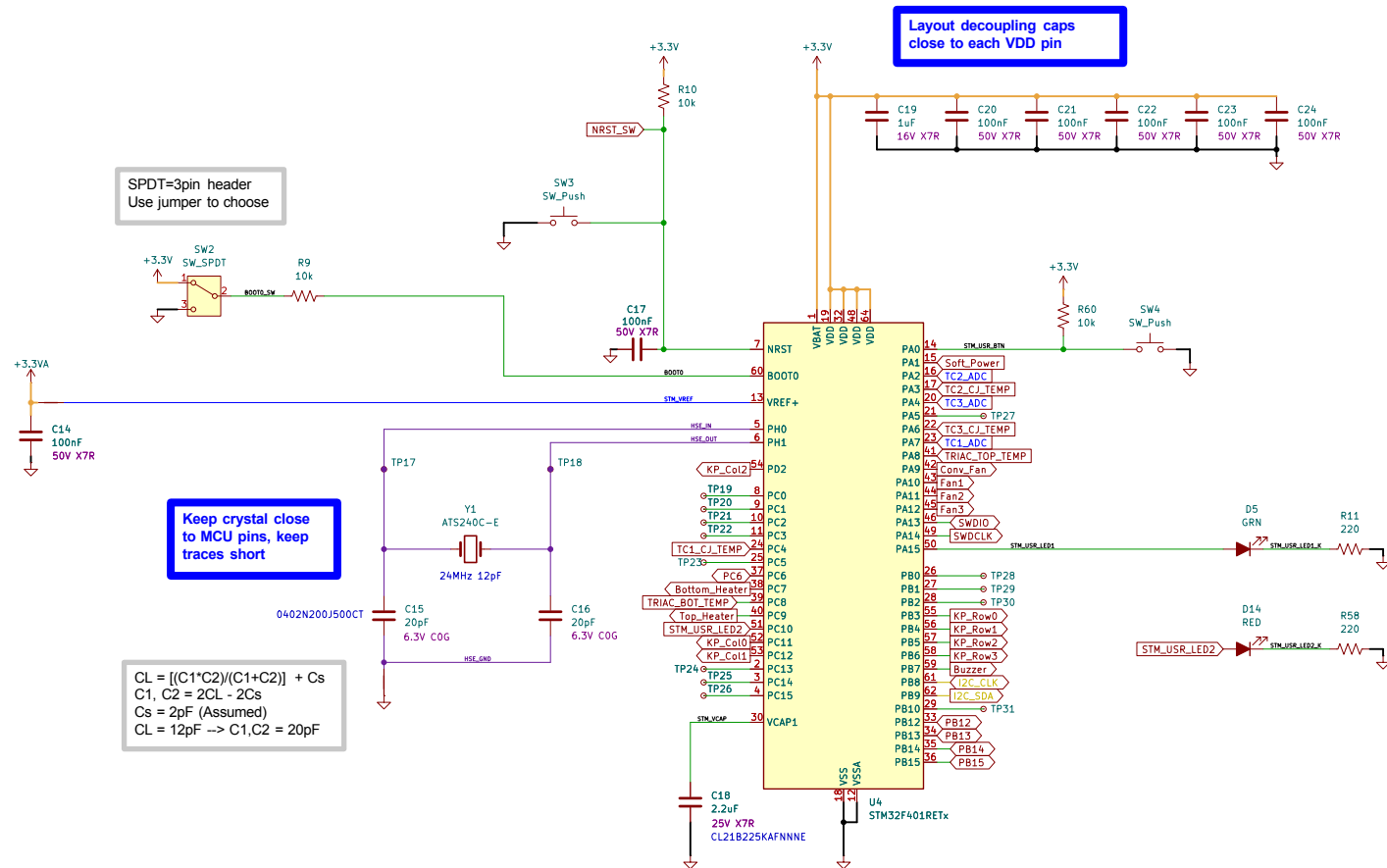


See MCP1727 datasheet typical application

PWRGD Max sink 1.2mA FET for LED, only on when PWRGD asserted

Comments:	Company:	Variant:	
IRM-10-5 Datasheet	MTP Engineering LLC		
MCP1727T-3302E/SN Datasheet	Board Name:	Project Name:	
	TF Main Board	ToasterFlo	
Sheet Title:	File Name:	Designer:	Date:
Power Supply	powerSupply.kicad_sch	Michael Pate	Last Modified Date
Sheet Path:	Reviewer:	Size:	Sheet:
/Project Architecture/Power Supply/	Michael Pate	A3	4 of 9

[5] Microcontroller



Comments: IRML6244TRPBF Datasheet STM32F401RET6 Datasheet	Company: MTP Engineering LLC		Variant:	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: Microcontroller	File Name: mcu.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/MCU/		Reviewer: Michael Pate	Size: A3	Sheet: 5 of 9

[6] TRIAC Output

DPST Switch
panel-mounted
to enclosure for
isolation

HIGH VOLTAGE:
Ensure appropriate isolation between AC and DC circuits!

HIGH POWER:
Ensure appropriate size traces for AC components!

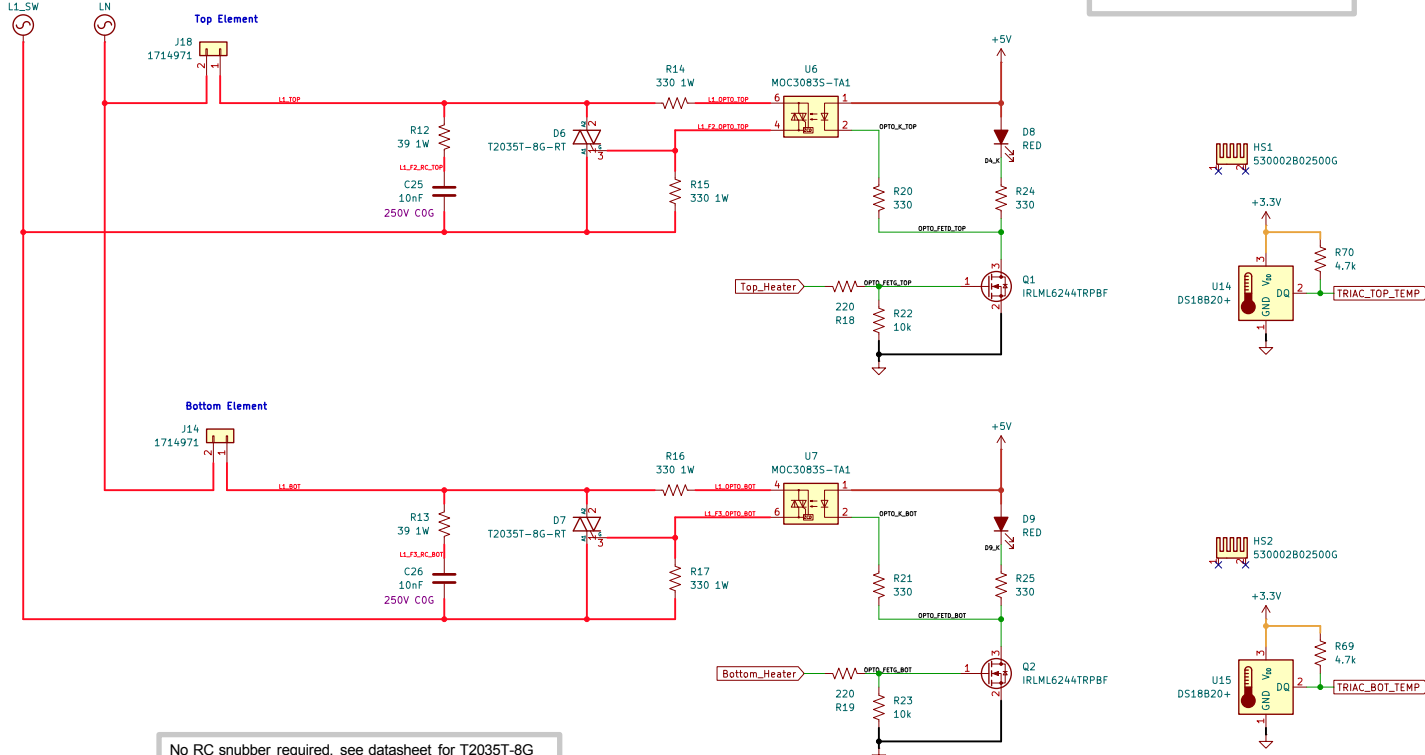
No RC snubber required, see datasheet for T2035T-8G
But for cheap insurance, use a 390/0.01uF RC snubber

At 60Hz, $X_{C2} \approx 265k\Omega$. At 120V, $I_{C2\text{RMS}} \approx 0.45mA$

Per MOC3085 Datasheet

$$V_{F,LED} = 1.4V @ I_F = 20mA$$
 $I_{FT}=5\text{mA}, I_{E,MAX}=50\text{mA}$

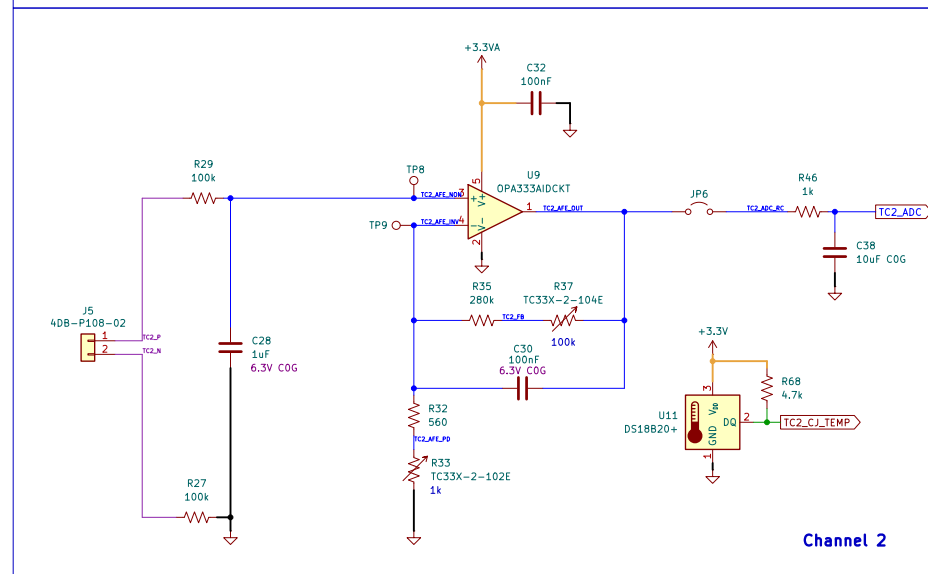
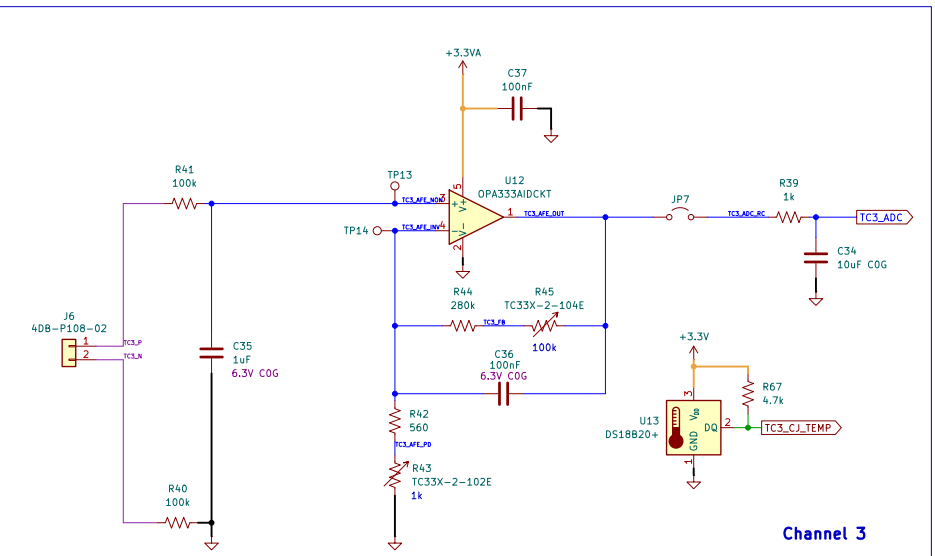
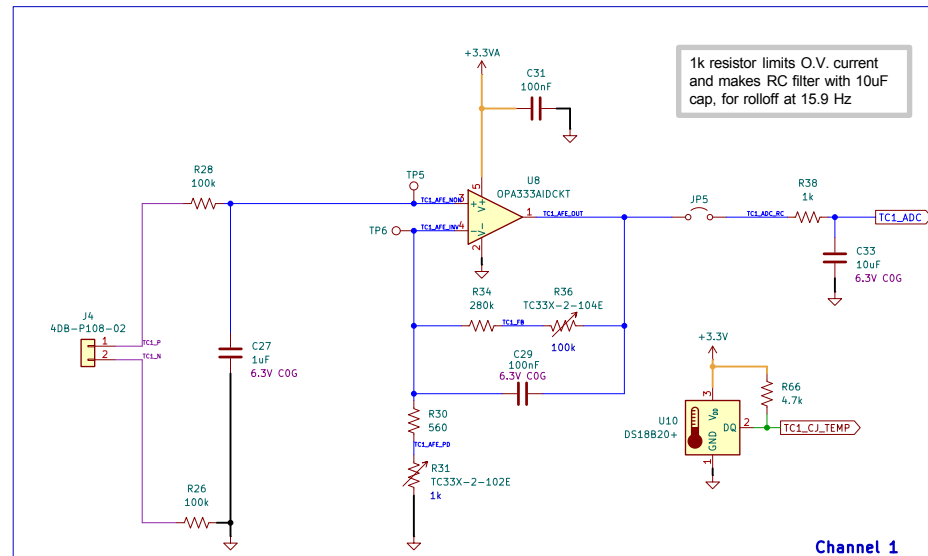
Use $R_5=330\Omega$ so $I_{LED} \approx 15mA$ for 5V IO from FET



DS18B20s mounted to HSs with thermal epoxy. Connected to board via JST XH

Comments: Link to TRIAC Circuit Example Link to T2025T Datasheet Link to MOC3083S Datasheet Link to Onsemi AN-3003 PDF CS867401080 Heatsink	Company: MTP Engineering LLC		Variant: 	
	Board Name: TF Main Board		Project Name: ToasterFlo	
Sheet Title: TRIAC Output	File Name: triac.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
Sheet Path: /Project Architecture/TRIAC/		Reviewer: Michael Pate	Size: A3	Sheet: 6 of 9

[7] Thermocouple Amplifiers



Use RCWCTE testpoint connectors on a Keystone 5019 pad

DS18B20 mounted near TC connector, for cold junction measurement/compensation

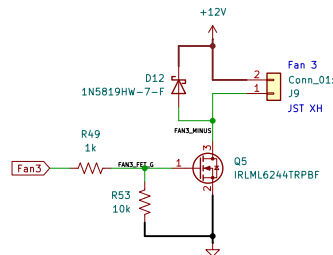
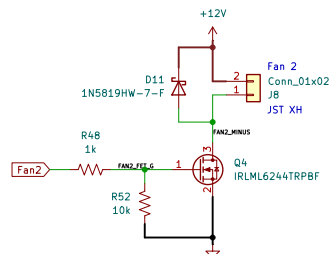
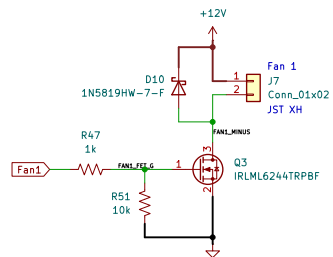
RC lowpass filter to remove most AC signals
 $R = 10\text{k}\Omega$, $C = 1\text{ }\mu\text{F}$
 $3\text{dB rolloff } f = (1 / (2\pi R C))$
 $f = (1 / (2\pi \cdot 10^4 \cdot 10^{-6})) = 15.92\text{Hz}$

C0G Capacitors instead of Type II ceramic to avoid microphony

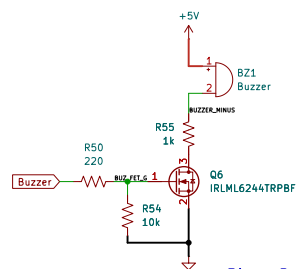
Layout RC filter near ADC inputs, as filter is designed to limit switching noise on quiet DC signals

	Comments: K Type TC Voltages OPA333AIDCKT Datasheet DS19820+ Datasheet		Company: MTP Engineering LLC Board Name: TF Main Board		Variant:	
	Sheet Title: Thermocouple Amplifiers		File Name: thermocouple.kicad_sch		Date: Last Modified Date	
	Sheet Path: /Project Architecture/Thermocouple/		Designer: Michael Pate		Revision: 2.0	
			Reviewer: Michael Pate		Size: A3	
					Sheet: 7 of 9	

[8] Interface I/O



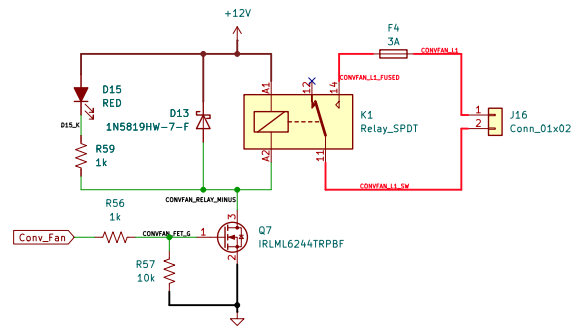
Enclosure Fans



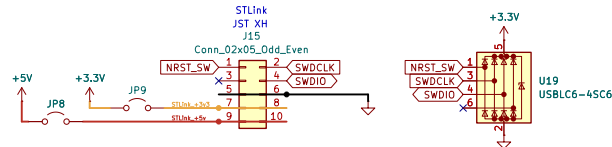
Piezo Buzzer

HIGH VOLTAGE:
Ensure
appropriate
isolation between
AC and DC
circuits!

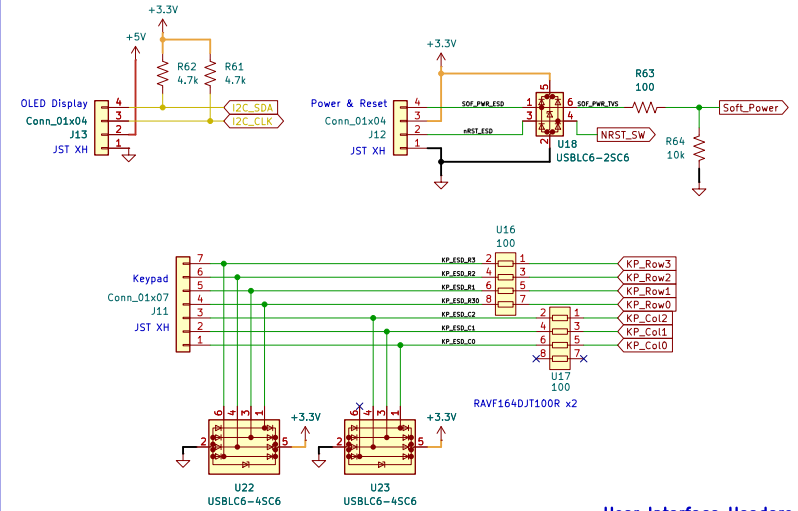
Connector for convection fan,
which is 120VAC powered. Relay
breaks line/neutral for the single
speed fan



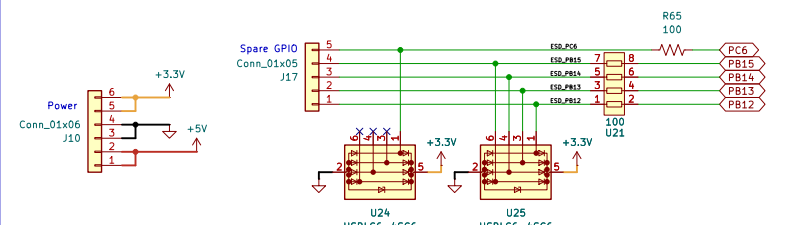
Convection Fan Control



STLink Programmer



User Interface Headers



Expandability Headers

Comments:	Company:	Variant:	
1N5819HW-7-F Datasheet	MTP Engineering LLC		
Sheet Title:	Board Name:	Project Name:	
Interface I/O	TF Main Board	ToasterFlo	
Sheet Path:	File Name:	Designer:	Date:
/Project Architecture/Peripherals and Connectors/	connectors.kicad_sch	Michael Pate	Last Modified Date
	Reviewer:	Size:	Sheet:
	Michael Pate	A3	8 of 9

[9] Revision History

DD.MM.YYYY - xxx Revision
Variant: xxx

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	Comments:	Company: MTP Engineering LLC		Variant:	
		Board Name: TF Main Board		Project Name: ToasterFlo	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Michael Pate	Date: Last Modified Date	Revision: 2.0
	Sheet Path: /Revision History/		Reviewer: Michael Pate	Size: A3	Sheet: 9 of 9